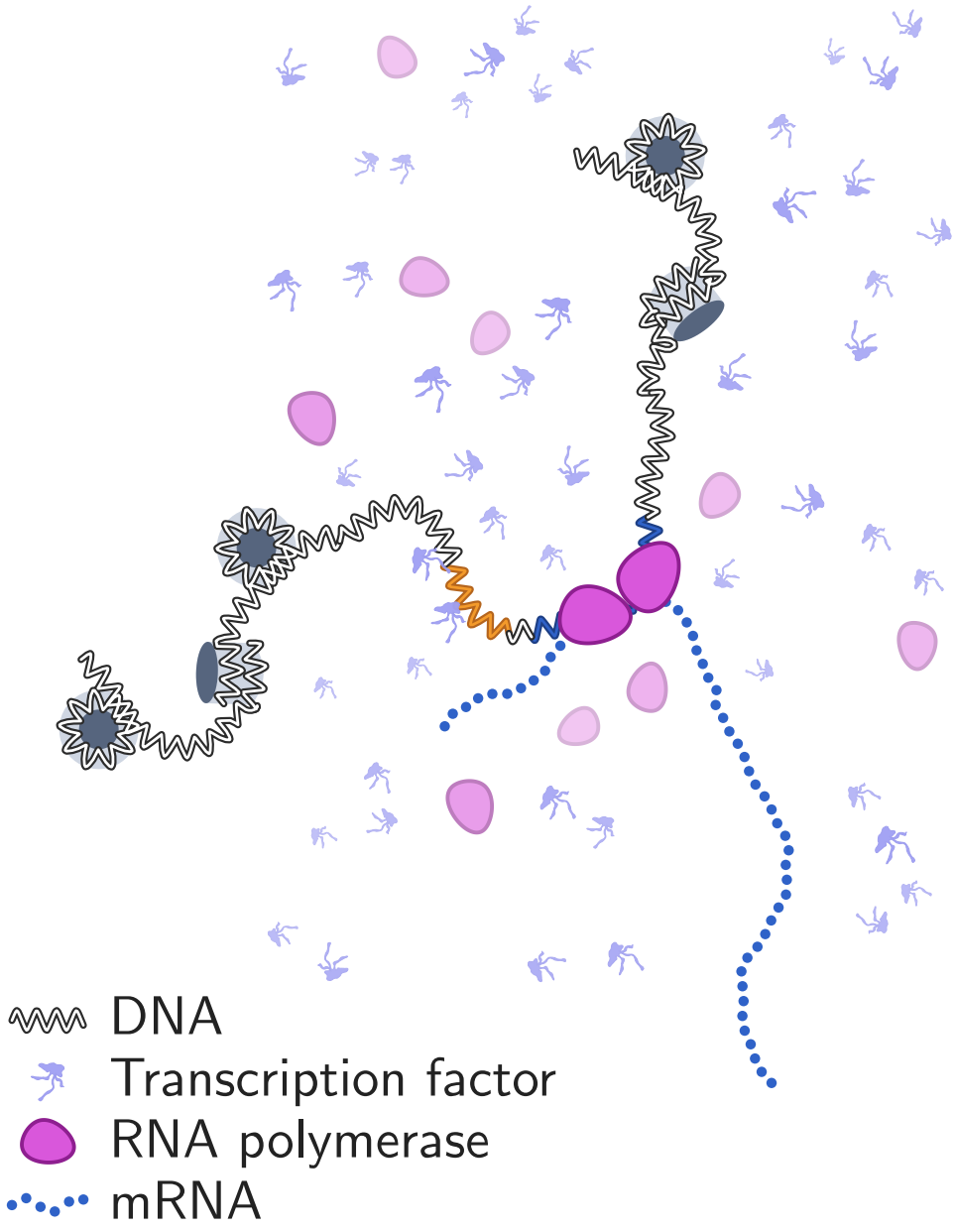
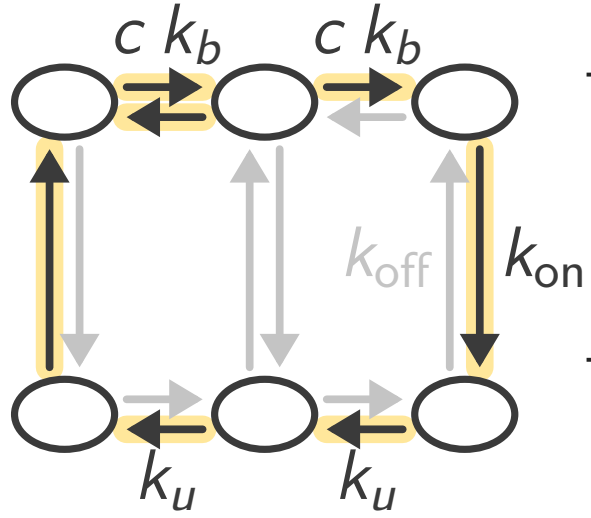
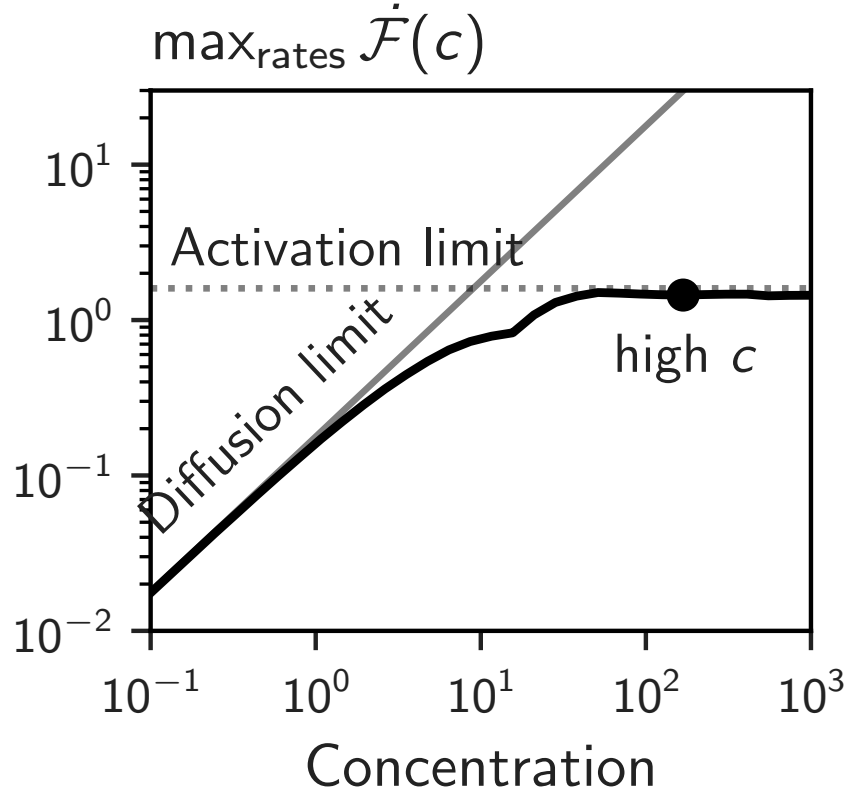

DNA
Transcription factor
RNA polymerase
mRNA


$c\,k_b$
$c\,k_b$
Transcription OFF
$k_{\text{off}}$
$k_{\text{on}}$
Transcription ON
$k_u$
$k_u$


$\max_{\text{rates}} \dot{\mathcal{F}}(c)$
$10^1$
$10^0$
$10^{-1}$
$10^{-2}$
Activation limit
Diffusion limit
high $c$
$10^{-1}$
$10^0$
$10^1$
$10^2$
$10^3$
Concentration